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Total Synthesis, Stereochemical Assignment and Divergent Enantioselective Enzymatic Recognition of Larreatricin

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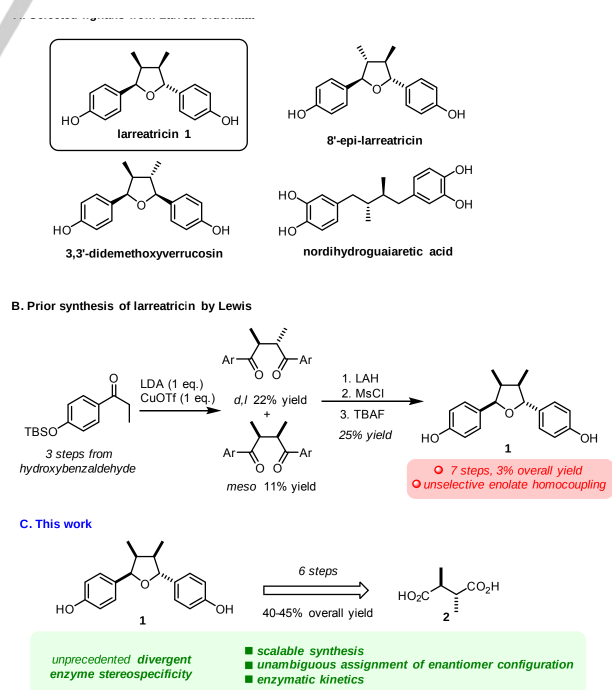
Abstract: We report a concise and efficient total synthesis of the lignan natural product larreatricin as well as an unambiguous assignment of configuration of its enantiomers, resolving a long-held controversy. Enzyme kinetic studies reveal that different polyphenol oxidases show high and remarkably divergent enantioselective recognition of this secondary metabolite.

Chirality is a ubiquitous feature of biological systems which plays a critical role in the metabolic processes of living organisms.[1] In particular, enzymatic reactions are highly stereospecific.[2] Polyphenol oxidases (PPO) are ubiquitous enzymes across a wide range of organisms from bacteria to fungi, plants and also mammals,[3-5] which contain a type III copper center in their active site.[6] Most PPOs are expressed as latent pro-enzymes *in vivo* containing a catalytically active domain and a C-terminal domain.[7] This C-terminal domain shields the PPO's active site and, as a result, the enzymes either possess only very weak or even no enzymatic activity unless they are activated by *e.g.* proteases, an acidic pH, fatty acids or detergents (e.g. SDS).[8-10] The precise physiological roles of polyphenol oxidases have hitherto been largely difficult to elucidate, especially since they accept a variety of aromatic substrates. This has led to considerable uncertainty about the roles played by these enzymes *in vivo*.[6] Tyrosinases (TYRs, EC 1.14.18.1 and EC 1.10.3.1), a subgroup of the PPO-family, use molecular oxygen to catalyze the *ortho*-hydroxylation of monophenols to catechols coupled with the subsequent oxidation of catechols to *o*-quinones.[4,5] The latter quinones are highly reactive and thus lead to the non-enzymatic formation of insoluble pigments like melanin. The best characterized PPO is the one from mushroom, where *Ab*PPO4[11] is reported to preferentially uptake *L*-tyrosine over *D*-tyrosine.[11a] The enantiospecificity of PPOs or the question of how PPOs select one enantiomer of a substrate over the other remain poorly investigated, open topics of high relevance for the understanding of how PPOs function *in vivo* and also how their enzymatic specificity may be tailored to provide an efficient tool for the enantioselective *o*-hydroxylation of a very broad range of phenolic compounds.

The naturally occurring lignan larreatricin **1** (Scheme 1A) belongs to a family of structurally related compounds occurring in the creosote bush *Larrea tridentata*.[12] This plant has a history of use in traditional South American medicine for various ailments including rheumatism, veneral diseases and digestive disorders, and other secondary metabolites derived thereof include the powerful antioxidant nordihydroguaiaretic acid (NDGA) cf. Scheme 1A. NDGA has potential applications in the treatment of a number of diseases, including various types of cancer, cardiovascular diseases and neurological disorders and is widely used for the treatment of actinic keratoses.[12] In 2003, Lewis reported that the enzyme (+)-larreatricin hydroxylase, isolated from *Larrea tridentata*, selectively takes up only the (+)-enantiomer of larreatricin as substrate.[12]

In the same year, the same authors reported a racemic synthesis of larreatricin using a low-yielding oxidative enolate homocoupling as key-step (Scheme 1B). Unfortunately, the poor overall yield of 3% for that synthesis over seven steps[13] as well as its lack of stereoselectivity (a mixture of all possible isomers was produced, with no apparent selectivity for the title compound) significantly hindered further investigation of this intriguing family of natural products. Additionally, an unambiguous assignment of the absolute configuration of naturally ocurring (-)-larreatricin is still missing after nearly three decades of research on this family of metabolites.[14]



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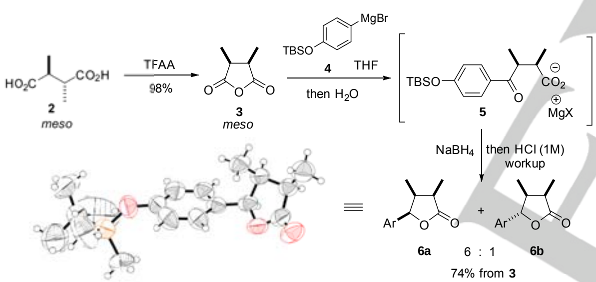
Scheme 1. (A) Lignan metabolites from *Larrea tridentate*, (B) prior low-yielding synthesis of **1** and (C) proposed scalable synthesis and enzymatic studies.

Given our prior interest in the stereoselective synthesis of tetrahydrofuran derivatives[15] and in the chemistry and biology of polyphenol oxidases[6] from a variety of plants[16][17][18] our collaborative interest was piqued by the enantiospecificity of larreatricin hydroxylase and the possibility that other tyrosinases might exhibit similar behavior. Herein, we report a concise, efficient and scalable enantioselective synthesis of **1**, an unambiguous structural assignment of its (+) and (–) enantiomeric forms as well as detailed kinetic studies of its divergent enantiospecific uptake by three different tyrosinases from two kingdoms (Scheme 1C) including (+)-larreatricin hydroxylase, which we were able to produce recombinantly for the first time.

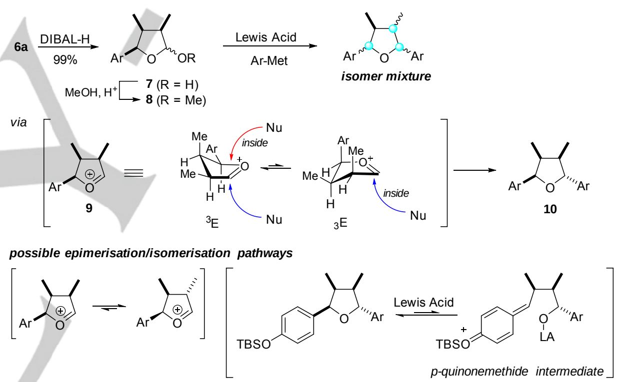
Larreatricin features some interesting symmetry elements. Of particular relevance for our retrosynthetic analysis are the vicinal methyl groups, which form a chirotopic *non*-stereogenic unit. Interestingly, the overall 2,3-diaryl-3,4-dimethyl-tetrahydrofuran core itself allows two possible *meso* configurations and four chiral arrangements (two each of C_1 and C_2 symmetry). These considerations inspired us to begin our synthesis from a *meso* precursor – in the event, the *meso* isomer of 2,3-dimethylsuccinic acid **2** (Scheme 2).



Scheme 2. Synthesis of stereodefined trisubstituted lactone **6a.** TFAA = trifluoroacetic anhydride. Ar = *p*-TBSO-C_6H_4.

This commercially available diacid was cyclodehydrated to the *meso* anhydride **3**[19] in nearly quantitative yield using trifluoroacetic anhydride (Scheme 2, top). The anhydride was then subjected to nucleophilic ring-opening with the Grignard reagent **4**, derived from silyl-protected 4-bromophenol. We quickly ascertained the lability of the ketoacid that is generated in this transformation, as all attempts to isolate it led to significant loss of material even though its formation appeared to proceed in high chemical yield. It was thus deemed advantageous to carry out *in situ* reduction of the magnesium carboxylate **5** with sodium borohydride followed by acidic (HCl) workup. This simple, scalable sequence led to a 6:1 diastereomeric mixture of trisubstituted lactones, of which all-*syn*-**6a** was the major component (Scheme 2). Isomer **6a** is a crystalline material and its structure and relative stereochemical assignment were secured by X-ray crystallographic analysis. Importantly, the overall sequence from diacid **2** to trisubstituted lactone **6a** requires a single chromatographic purification, delivering pure material in 62% overall yield for the 3 steps in multi-gram scale.

At this stage, we envisaged a lactone partial reduction (easily achieved in quantitative yield using diisobutylaluminium hydride; Scheme 3) followed by substitution of the lactol **7**/acetal **8** with the second aryl nucleophilic equivalent. However, several attempts employing a range of Lewis acids and organometallic nucleophiles delivered not only **1** (< 5%) but also multiple isomeric natural product congeners (such as, *e.g.*, those shown in Scheme 1A) at an overall modest yield. If we assume that the reaction proceeds via the oxacarbenium ion **9**, the preferred trajectory for attack of the incoming nucleophile should be *anti* to the C-3 methyl group for both conformers and thus deliver the desired configuration. The formation of multiple isomers instead suggests that either (a) the oxacarbenium **9** quickly epimerises (via the enol ether) competitively with nucleophilic addition or (b) the tetrahydrofuran **10** suffers further erosion of configurational purity and/or destruction initiated by Lewis acid mediated ring-opening *p*-quinone methide formation.[20]



Scheme 3. Unsuccessful attempts at installing the fourth substituent. DiBAL-H = diisobutylaluminum hydride.

The need to prevent epimerization and side reactions suggested structurally modifying the leaving group at carbon C2, in order to avoid the need for a strong Lewis acid. After some experimentation (Scheme 4), we found that conversion of lactol **7** to a sulfone acetal **11**[21] followed by addition of the Grignard reagent **4** in the presence of zinc (II) bromide gratifyingly generated tetrasubstituted tetrahydrofuran **10**[13] as the main diastereomeric species in 70-80% yield from lactol **7**.[22] Desilylation finally delivered *rac*-**1** in six steps and an overall yield of 37-45% from commercially available succinic acid **2**, a significant improvement in efficiency and practicality from the prior reports.[13] Chiral HPLC separation provided samples of both (-) and (+)-larreatricin, as it was important to obtain and characterize both enantiomers for subsequent structural and enzymatic studies (*vide infra*).

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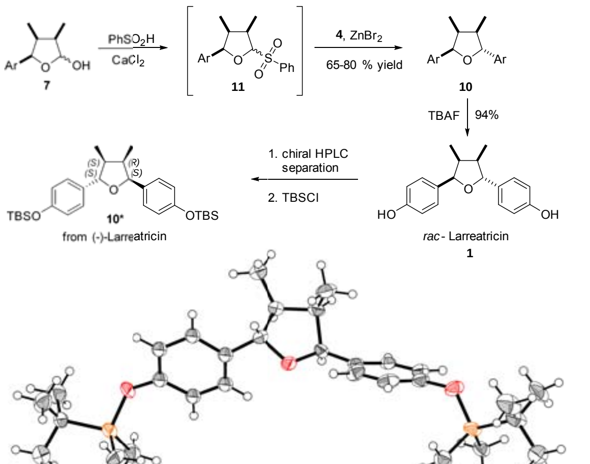
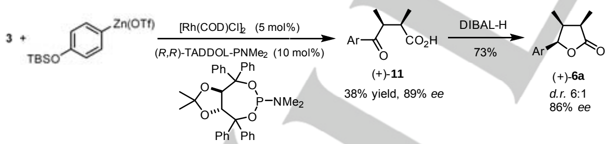


10*

Scheme 4. Completion of the synthesis of **1** and assignment of the absolute configuration of (-)-**1** by X-ray crystallography. Ar = *p*-TBSO-C_6H_5 TBAF = tetrabutylammonium fluoride.

Importantly, the separation of enantiomers also enabled us to unambiguously establish the absolute configuration of (-)-larreatricin through X-ray crystallographic analysis of the silylated derivative **10***. As shown in Scheme 4, (-)-larreatricin has a (2*S*, 5*S*)-configuration. This is, to the best of our knowledge, the first unambiguous stereochemical assignment of this natural product.

In parallel, we developed an enantioselective synthesis of the key lactone (+)-**6a** (Scheme 5) using an anhydride desymmetrization approach pioneered by Rovis.[23] *In situ* reduction of the carboxylate (as before) was not possible under this protocol, and the aforementioned instability of (+)-**12** is responsible for its modest isolated yield. Notably, ketone reduction in this sequence is also very sensitive to reaction conditions giving access to both diastereomers.[24] Nonetheless, reduction of (+)-**12** with DIBAL-H afforded (+)-**6a** in high enantioselectivity and comparable diastereoselectivity to the racemic pathway. This thus establishes an enantioselective route for the total synthesis of (+)-larreatricin.



Scheme 5. Enantioselective desymmetrisation of **3** en route to (+)-**1**.

At this juncture, we investigated the enantioselectivity towards the substrate larreatricin for three recombinantly expressed PPOs: the only so far described enantiospecific plant enzyme larreatricin hydroxylase from *Larrea tridentata* (*Lt*PPO)[12, this work], for which we established a heterologous production protocol yielding approximately 20 mg od protein per l of bacterial culture, the apple tyrosinase (*Md*PPO1)[16b] and fungal PPO from *Agaricus bisporus* (*Ab*PPO4)[11a] for which also the proteolytically activated form (*Ab*PPO4-act)[11a] was investigated (SI, Figure S4). This activated form is derived from *Ab*PPO4 by removal of the C-terminal domain which relieves the latency of the PPO[11a] and therefore allows for the direct comparison of enzymatic activity liberated by treatment with a detergent to the enzymatic reaction carried out in the absence of any external activator.

The enzymatic *o*-hydroxylation and oxidation of phenolic compounds by PPOs gives rise to reactive *o*-quinones which do usually evolve to dark pigments of the melanin family.[25] As this reactions can complicate the kinetic assay we applied the potent nucleophile 3-methyl-2-benzothiazolinone hydrazone (MBTH) as a chinone-trapping agent that yields soluble and reasonably stable coupling products.[26] The three enzymes revealed completely different enantiospecificities for (+) and (-)-larreatricin (cf. Table 1). The lag-Phase usually encountered when monitoring the PPO-reaction on monophenolic substrates[27] was shorter than two seconds for the preferred enantiomer of larreatricin for each of the four tested enzyme preparations. For the non-preferred substrates it took up to one minute until the steady-state rate of the reaction was reached, substrate conversion at this point did not exceed 15 % of the initial substrate concentration.

Table 1. Kinetic characterization of the reaction of larreatricin with three PPOs

Enzyme	Larreatricin	K_M / mM	k_{cat} / s^{-1}
*Lt*PPO [this work]	(+)	0.062 ± 0.0044	11.3 ± 0.58
	(-)	0.27 ± 0.020	0.48 ± 0.02
*Md*PPO1[16b]	(+)	0.21 ± 0.019	0.61 ± 0.02
	(-)	0.34 ± 0.031	0.34 ± 0.02
*Ab*PPO4[11a]	(+)	0.25 ± 0.036	1.14 ± 0.12
	(-)	0.28 ± 0.025	14.9 ± 0.93
*Ab*PPO4-act[11a]	(+)	0.11 ± 0.010	0.91 ± 0.033
	(-)	0.31 ± 0.072	12.1 ± 1.8

Volumetric activity was determined by following the appearance of the quinone adduct, measured photometrically at λ_{max} = 501 nm, $\varepsilon(\lambda_{max})$ = 37 mM^{-1} cm^{-1}.

*Lt*PPO displayed a clear preference for (+)-larreatricin and the k_{cat} value was 24-times higher than for the reaction with (-)-larreatricin (k_{cat} = 11.3 s^{-1} and 0.48 s^{-1}, respectively). This affirms the previously observed enantiospecificity of the enzyme[12], but our results show that (-)-**1** is also accepted as a substrate by *Lt*PPO. The preference of *Lt*PPO for (+)-larreatricin over the (-)-enantiomer clearly indicates that the latter is the natural substrate of *Lt*PPO. The K_M values for (+) and (-)-**1** are 0.062 mM and 0.27 mM, respectively and the 62 µM for (+)-larreatricin is among the lowest K_M-values reported for PPOs, pointing towards a very strong enzyme-substrate interaction.

The second plant enzyme *Md*PPO1 shows no clear preference for either one of the enantiomers with the k_{cat} on (+)-larreatricin being 0.61 s^{-1}, which is only 1.8-times higher than the k_{cat} for the (-)-enantiomer (k_{cat} = 0.34 s^{-1}). The specificity of *Md*PPO1 is also similar for the two enantiomers, with the K_M value for (-)-larreatricin (0.34 mM) being a little higher than for (+)-larreatricin (K_M = 0.21 mM).

Surprisingly, the mushroom enzyme *Ab*PPO4 exhibits a reversed specificity for the two enantiomers of larreatricin. It reacts on (-)-larreatricin (k_{cat} = 15 s^{-1}) 13 times faster than on (+)-larreatricin (k_{cat} = 1.1 s^{-1}). The K_M-value was found to be 0.25 mM for (+)-larreatricin and 0.28 mM for (-)-larreatricin, which indicates a very similar specificity. In the case of *Ab*PPO4 we compared the findings for the latent enzyme with the active enzyme (termed *Ab*PPO4-act)[11a] which did show very similar kinetic parameters and was also 13 times faster on the (+)-enantiomer (Table 1).

These enantioselectivies are among the top-rank of stereoselectivity reported for PPOs so far. For the most well-studied system, tyrosinase isolated from mushroom (*Agaricus bisporus*) no significant preference for either *L*- or *D*-tyrosine was found.[28] A slight preference for *L*-DOPA was displayed by tyrosinase from the bacterium *Bacillus megaterium* which reacted 1.7-times faster on the S-enantiomer than on R-DOPA.[29] Another bacterial tyrosinase from *Ralstonia solanacearum* showed a medium enantiopreference for *L*-tyrosine (4.7 times faster than on the R-enantiomer) which could be reduced to almost no selectivity (0.98, i.e. a very slight preference for the R-enantiomer) by mutation of one amino acid far from the active site.[30] Enantioselectivities similar to the values presented herein for larreatricin were seen in a secreted fungal tyrosinase from *Trichoderma reesei* that reacted 14.3 times faster on 2.5 mM *L*-tyrosine than on the R-enantiomer of this amino acid (based on specific activities as no kinetic characterization is given).[31] The highest enantioselectivity reported for a PPO comes from tyrosinase produced by the bacterium *Streptomyces sp.* REN-21 that displayed resistence to organic solvents and was 35.9 times faster on *L*- than on *D*-tyrosine.[32] Comparison of our results with those obtained using those three bacterial and two fungal tyrosinases and especially the increase in apparent stereoselectivity of *Ab*PPO4 when going from the simple amino acid tyrosine (2.6-fold faster on the S-enantiomer) to the lignan larreatricine (13 times faster reaction on the (-)-enantiomer) indicate that the enantioselectivity in the reaction with larreatricin is determined to a large extent by the special geometry of this rather large substrate. For *Lt*PPO the preferred enantiomer of larreatricin also exhibits the smaller K_M value as would be expected for tighter binding. While the difference in K_M is small to negligible for *Md*PPO1 and *Ab*PPO4 a different trend is noted: The enantiomer that is catalysed faster has the higher (while still low) K_M value which usually indicates less favorable binding to the active site. This hints towards a second factor determining the reaction rate besides the accurate fit of the substrate to the active site of the enzyme. The notable difference in the preference for the (+)-enantiomer exhibited by the enzymes *Lt*PPO and *Md*PPO1 on the one hand and for the (-)-enantiomer by *Ab*PPO4 on the other hand gives a clear indication that the active center of PPOs is able to discriminate between enantiomeric substrates. The generally low K_M values point towards a strongly bound substrate.

In conclusion, we have reported a concise and efficient total synthesis of the lignan natural product larreatricin as well as an unambiguous assignment of the absolute configuration of its enantiomers. Our synthesis also enabled detailed enzyme kinetic studies with different classes of polyphenol oxidases revealing high and remarkably divergent enantioselectivity in their recognition of this natural product, providing an intriguing basis for further studies of the enzyme-substrate binding mode in the PPO family.

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Keywords: polyphenol oxidase • total synthesis • enantiospecific • enzyme kinetics • lignan • larreatricin

[1] a) R. Bentley in *Reviews in Cell Biology and molecular Medicine*, Wiley, 2006, pp. 579–618; b) D. G. Blackmond, *Cold Spring. Harb. Perspect. Biol.* **2010**, *2*, 1-17; c) L. A. Nguyen, H. He, C. Pham-Huy, *Int. J. Biomed. Sci.* **2006**, *2*, 85-100.
[2] See, e.g.: L. Zemanova, P. Solich, V. Wsól, *Curr. Drug Metabol.* **2010**, *11*, 547-559.
[3] M. Pretzler, A. Bijelic, A. Rompel in *Reference Module in Chemistry, Molecular Sciences and Chemical Engineering* (Elsevier, 2015).
[4] R. Malkin, B. G. Malmström, in *Advances in Enzymology and Related Areas of Molecular Biology* (ed. Nord, F. F.) pp. 177–244 (John Wiley & Sons, Inc., 1970).
[5] B. G. Malmstrom, *Ann. Rev. Biochem.* **1982**, *51*, 21–59.
[6] M. Pretzler, A. Rompel, *Inorg. Chim. Acta* **2018**, doi: 10.1016/j.ica.2017.04.041.
[7] L.T. Tran, J.S. Taylor, C.P. Constabel, *BMC Genomics* **2012**, *13*, 395.
[8] R. Yoruk, M.R. Marshall, *J. Food Biochem*. **2003**, *27*, 361-422.
[9] D. Nillius, E. Jaenicke, H. Decker, *FEBS Letters* **2008**, *582*, 749-754.
[10] N. Fujieda, S. Itoh, *Bull. Chem. Soc. J.* **2016**, *89*, 733-742.
[11] a) M. Pretzler, A. Bijelic, A. Rompel, *Sci. Rep.* **2017**, 7, srep1810; b) S. G. Mauracher, C. Molitor, R. Al -Oweini, U. Kortz, A. Rompel, *Acta Crystallogr., Sect. D: Biol. Crystallogr.* **2014**, *70*, 2301-2315; c) S. G. Mauracher, C. Molitor, C. Michael, M. Kragl, A. Rizzi, A. Rompel, *Phytochemistry* **2014**, *99*, 14-25.
[12] M.-H. Cho, S. G. A. Moinuddin, G. L. Helms, S. Hishiyama, D. Eichinger, L. B. Davin, N. G. Lewis, *Proc. Natl. Acad. Sci. U. S. A.* **2003**, *100*, 10641–10646.
[13] N. G. Lewis, S. G. A. Moinuddin, S. Hishiyama, M. Cho, L. B. Davis, *Org. Biomol. Chem*. **2003**, *1*, 2307-2313.
[14] The absolute configuration larreatricin enantiomers remains, to date, an unsolved problem. For an indirect assignment of 8'-*epi*-larreatricin and its 3'-OH analogue, see: C. Konno, Z.-Z.Lu, H.-Z. Xue, C.A.J. Erdelmeier, D. Meksurien, C.-T. Che, G.A. Cordell, D.S. Soejarto, D.P. Waller and H.H.S. Fong, *J. Nat. Prod.* **1990**, *53*, 396.
[15] J. Sabbatani, N. Maulide, *Angew. Chem. Int. Ed.* **2016**, *55*, 6780-6783.
[16] a) F. Zekiri, C. Molitor, S.G. Mauracher, C. Michael, R. L. Mayer, C. Gerner, A. Rompel, *Phytochemistry* **2014**, *101*, 5-15; b) I. Kampatsikas, A. Bijelic, M. Pretzler, A. Rompel, *Sci. Rep.* **2017**, *7*, srep8860.
[17] A. Bijelic, M. Pretzler, C. Molitor, F. Zekiri, A. Rompel, *Angew. Chem., Int. Ed.* **2015**, 54, 14677-14680; *Angew. Chem.* **2015**, 127, 14889-14893.
[18] C. Molitor, S.G. Mauracher, A. Rompel, *Proc. Natl. Acad. Sci. USA* **2016**, *113,* E1806-E1815.
[19] a) D. Marcoux, P. Bindschädler, A. W. H. Speed, A. Chiu, J. E. Pero, G. A. Borg, D. A. Evans, *Org. Lett.* **2011**, *13*, 3758-3761; b) J. K. Crandall, T. Schuster, *J. Org. Chem*. **1990**, *55*, 1973-1975.
[20] For related observations, see: K. Harada, M. Kubo, H. Horiuchi, A. Ishii, T. Esumi, H. Hioki, and Y. Fukuyama, *J. Org. Chem*. **2015**, *80*, 7076–7088.
[21] H. Kim, A. C. Kasper, E. J. Moon, Y. Park, C. M. Wooten, M. W. Dewhirst, J, Honge, *Org. Lett.* 2009, **11**, 89-92.
[22] Yield fluctuation is likely due to the sensitive nature of the intermediates towards the reaction conditions and reagent purity.
[23] a) J. B. Johnson, E. A. Bercot, C. M. Williams, T. Rovis, *Angew. Chem. Int. Ed.* **2007**, *46*, 4514-4518; *Angew. Chem.* **2007**, *119*, 4598-4602. b) J. B. Johnson, M. J. Cook, T. Rovis, *Tetrahedron* **2009**, *65*, 3202-3210.

10.1002/chem.201803785

[24] Indeed, various solvents, additives and reagents allow diastereoselectivities ranging from 6:1 to 1:10. See the SI for details of the screening of reaction conditions.

[25] S. Ito, *Pigm. Cell Res.* **2003**, *16*, 230-236.

[26] A. J. Winder, H. Harris, *Eur. J. Biochem.* **1991**, *198*, 317-326.

[27] J. Ros, J. Rodríguez-López, F. García-Cánovas, *Biochim. Biophys. Acta (BBA) - Protein Structure and Molecular Enzymology* **1994**, *1204*, 33-42.

[28] J. C. Espín, P. A. García-Ruiz, J. Tudela, F. García-Cánovas, *Biochem. J.* **1998**, *331*, 547-551

[29] V. Shuster, A. Fishman, *J. Mol. Microbiol. Biotechnol.* **2009**, *17*, 188-200.

[30] S. Molloy, J. Nikodinovic-Runic, L. B. Martin, H. Hartmann, F. Solano, H. Decker, K. E. O'Connor, *Biotechnol. Bioeng.* **2013**, *110*, 1849-1857.

[31] E. Selinheimo, M. Saloheimo, E. Ahola, A. Westerholm-Parvinen, N. Kalkkinen, J. Buchert, K. Kruus, *FEBS Journal* **2006**, *273*, 4322-4335.

[32] M. Ito, K. Oda, *Biosci. Biotechnol. Biochem.* **2000**, *64*, 261-267.

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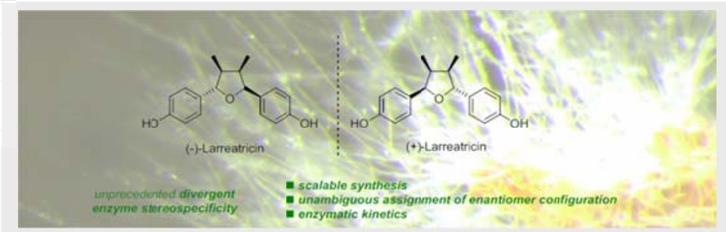
COMMUNICATION



Recognise your mirror image: We report a concise and efficient total synthesis of the lignan natural product larreatricin as well as an unambiguous assignment of configuration of its enantiomers. Enzyme kinetic studies reveal that different families of polyphenol oxidases show high and remarkably divergent enantioselectivity in their recognition of this secondary metabolite.

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